

SOFTWARE ENGINEERING LAB-1 | SHRIYADITA MOHANTY | PES1UG24AM270 | AIML-E

PROBLEM STATEMENT #61 — Household Carbon Footprint & Target Tracker

1. Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO ₂ e) emissions.	High	Pass: Calculated emissions match the specified EPA conversion-factor formulas within an allowed rounding tolerance. Fail: Emissions are zero/negative or calculated using incorrect factors.	Provides the core carbon-footprint calculation needed to measure household emissions.
FR-002	Functional	The system shall allow residents to record and update monthly electricity, gas, and commute-distance consumption data.	High	Pass: A resident can successfully add, edit, and save valid monthly consumption records. Fail: Valid records cannot be saved or contain missing required values after submission.	Enables households to maintain accurate and up-to-date consumption information.
FR-003	Functional	The system shall display monthly and annual CO ₂ e emissions and historical trends based on recorded household consumption.	High	Pass: The system displays calculated monthly and annual emissions and a historical trend for all months with available data. Fail: Displayed values do not match stored calculations.	Helps residents understand their emissions patterns over time.
FR-004	Functional	The system shall allow residents to set an annual carbon-reduction target and track progress toward the target.	High	Pass: The system stores the selected target and displays current reduction progress as a percentage and/or chart. Fail: Target or progress is not correctly stored or calculated.	Allows households to establish measurable sustainability goals.
FR-005	Functional	The system shall allow the Sustainability Coach to review a household's carbon-emission history and provide reduction recommendations.	Medium	Pass: The Sustainability Coach can access the household's historical emissions and record at least one recommendation. Fail: The coach cannot access the required history or save a recommendation.	Provides expert guidance to help households achieve their carbon-reduction goals.
NFR-001	Non-Functional	The annual carbon reduction milestone chart must render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking tests confirm chart rendering within 200 ms under the defined simulated peak load while meeting required security standards. Fail: Rendering exceeds 200 ms or security requirements are violated.	Ensures the tracking interface remains responsive while protecting household data.
NFR-002	Non-Functional	The system shall provide a clear and user-friendly interface for entering consumption data and shall preserve successfully submitted records without data loss.	Medium	Pass: At least 90% of test users can complete a monthly consumption entry without assistance, and 100% of successfully submitted records remain retrievable. Fail: Fewer than 90% complete the task or any successfully submitted record is lost.	Makes regular household data entry simple and ensures users can rely on their stored records.

3. Use-Case Flow Specification

Use Case: Compute Monthly Carbon Footprint (CO₂e Emissions)

Use Case ID	UC-01	Related Requirement	FR-001
Primary Actor	Resident	Secondary Actor	System (Emissions Calculation Engine)
Trigger	Resident submits monthly electricity and transport consumption data and requests calculation of the carbon footprint for that month.		

Preconditions

- 1. The Resident is registered and successfully logged into the system.
- 2. The Resident has already recorded (or is recording) at least one valid monthly electricity (kWh) and transport (km) log for the target month (FR-002).
- 3. The system's EPA conversion-factor formulas are configured and available.

Postconditions

Success: The total monthly CO₂e emission value is calculated, stored against the household and the selected month, and displayed to the Resident along with the updated historical trend (FR-003).

Failure: No emission record is created or updated; the system state remains unchanged from before the use case began, and the Resident is informed of the reason for failure.

Main Success Scenario

- 1. The Resident navigates to the “Monthly Consumption” screen and selects the month to calculate.
- 2. The system retrieves the Resident's saved electricity (kWh) and transport (km) logs for the selected month (includes UC: *View Consumption Records*).
- 3. The Resident confirms the displayed consumption values and selects “Calculate Emissions.”
- 4. The system applies the EPA conversion-factor formulas to the electricity and transport values to compute the total CO₂e emissions for the month.
- 5. The system validates that the calculated emission value is a positive, non-zero number within expected rounding tolerance.
- 6. The system stores the calculated CO₂e value against the household record for the selected month.
- 7. The system updates and displays the monthly and annual emissions summary along with the historical trend chart (FR-003).
- 8. The system checks the Resident's active reduction target and updates the progress indicator (includes UC: *Track Reduction Target*, FR-004).

Alternate Flow — A1: Missing or Invalid Consumption Data

Triggered at step 3, when required consumption data for the selected month is incomplete or invalid.

- 3a. The system detects that electricity or transport values are missing, negative, or non-numeric for the selected month.
- 3b. The system displays an error message identifying the missing or invalid field(s) and blocks the calculation.
- 3c. The Resident is prompted to add or correct the consumption record (extends UC: *Record Consumption Data*, FR-002).
- 3d. Once the Resident provides valid data and resubmits, the flow resumes at step 4 of the Main Success Scenario.

Exception Flow — E1: Calculation Engine Failure

If the emissions calculation engine is unavailable or an internal computation error occurs at step 4, the system aborts the calculation, leaves any previously stored emission value unchanged, and displays a retry message to the Resident without recording a new (zero or invalid) emission value.